Toasting

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Introduction

Toasting is a vital component in the manufacture of breakfast cereals. Consumers find many of the attributes, such as flavor, color, and texture, developed during the toasting process most appealing.

The tradition of using toasting to differentiate items began with bakers, who utilized specialized ovens to create new breads and pastries. Cereal makers adapted this initial technology and now almost all cereal manufacturers employ toasting to enhance flavor development and differentiate brands.

In cereal toasting processes, the product moves sequentially through heating and drying phases to reach the conditions required to kick off the Maillard browning reactions and caramelization required for flavor and color development. Once activated, the product must absorb enough energy to continue the reactions.

In production, it is hard to measure the surface temperature of a product and even more difficult to manage the Maillard browning reactions. One method to estimate the start of the Maillard browning reactions is to determine how much energy is available. Sugars start to react with proteins when the product surface reaches 248−284°F (120−140°C) (Reineccius, 2006). Before reaching activation temperatures, energy absorbed by the product is also utilized to heat the dry mass, heat the water remaining around the dry mass, and evaporate water from the dry mass. The remaining energy can be utilized for the Maillard browning reactions. Evaporation removes mass and with the mass leaves energy; the net result is the cooling of the product surface. High rates of evaporation keep the product surface cool and prevent reactions from starting.

The product texture changes when the structural properties of the starch matrix are modified by expansion or shrinkage. Physical changes in the starch matrix also require energy. Structural changes affected by moisture removal will affect the glass transition temperature (T_g) of the matrix, which will influence diffusivity and rates of reaction (see Chapter 6). Several methods, such as thermal analysis, can be used to measure T_g.

The product leaving the toasting oven must have uniform color. Variation in color is considered a quality defect. Uniform product color requires control when the Maillard browning reactions start and when they stop; therefore, toasting ovens require a higher

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standard for uniformity than other drying operations. Being able to control chemical reactions is critical to the final product color and flavor. Research on reactions that control flavor and color development continues in academia and the food industry. To survey the art of cereal toasting, one must first be able to reference and utilize relevant basic food science and engineering principles on toasting.

Toasting principles

The basic changes occurring in the product as it passes through an oven are as follows:

- Increase in product temperature
- Decrease in product moisture content
- Change in product density (shrinkage or expansion)

Chemical reactions, such as Maillard browning reactions, lead to flavor and color development. The rise in product temperature and moisture reduction can be measured by taking and analyzing actual sample temperatures and moisture contents. The change in product density, due to expansion or shrinkage, can also be measured directly. The increase in product flavor or sweetness is difficult to measure directly as it goes through toasting. However, it can be estimated using the product temperature increase as a guideline as discussed below.

The classical drying curve plots product moisture content versus time. Plotting the product temperature versus time gives an insight on product changes during toasting (Fig. 15.1). The toasting operation can be divided into four phases: heat up, drying, expansion, and color and flavor development. The phases describe how the product changes in the oven. All cereal products utilize three of the phases, the expansion phase is added for puffed products. The four-phase toasting process is based on the concept of critical moisture content. Critical moisture content is the point at which control of the drying process changes from being energy limited to moisture migration limited.

The first phase shows the product heating up until it reaches an equilibrium temperature, which marks the start of the second phase, where energy absorbed from the air is balanced by the energy removed by evaporation. The second phase ends when the product reaches its critical moisture content, at which point the evaporation slows down and the product temperature rises quickly. The third phase starts with a steep rise in product temperature, which quickly exceeds the boiling point leading to the fourth phase, expansion. The liquid water trapped inside the product vaporizes, leading the product to expand and/or puff. After this expansion, temperature increases and chemical reactions involved in flavor and color development start.

Breaking the toasting process down into phases allows one to focus on how the product is changing stepwise during toasting. The graphs in Figs. 15.2–15.6 detail changes in energy when a cereal flake product undergoes puffing and toasting.

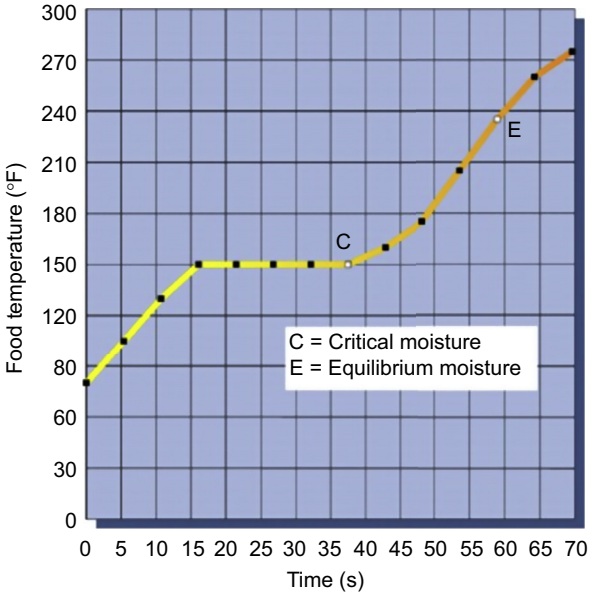

Figure 15.1 Product temperature versus time: the curve plots product moisture content versus time, which provides insight into how the product changes in the toasting process.

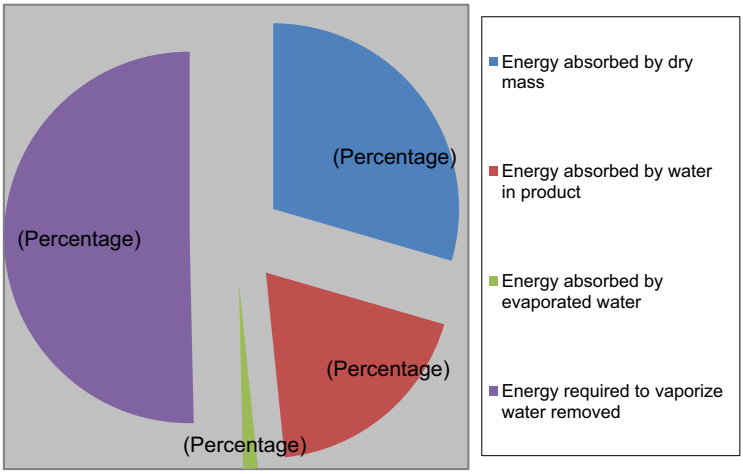

Figure 15.2 Heating phase: energy absorbed by product.

For operators, the information can be used to troubleshoot production issues. For the designers, the information can be used to improve the toasting application and equipment. This understanding can then be utilized to improve oven operation and oven component design.

Heating phase

In the first step of the toasting process, flakes undergo a heating phase, in which the large amount of the energy absorbed increases the temperature of the dry mass and liquid water in the flake. In Fig. 15.2, the blue area represents the energy absorbed by the dry mass. The red area represents the energy absorbed by the liquid water. In the heating phase, the energy absorbed by the dry mass and the water represents 49% of the total energy absorbed. The purple area represents the energy used in evaporation of water.

Constant rate and falling rate drying phase

The second event, the constant rate drying phase, occurs when the product temperature rise slows down and the evaporation rate increases, as shown in Fig. 15.3. In this section, product surfaces are being cooled by water evaporating away from the flake surface. The energy absorbed by the dry mass and the energy absorbed by the internal water drop to 26% of the total energy absorbed. The amount of water evaporated is only limited by energy absorbed; the constant rate phase ends when the flake moisture falls below the critical moisture content.

The falling rate drying phase starts when flakes reach critical moisture content, which is defined as the point at which moisture moving to the surface is slower than moisture evaporating from the flake surface (Geankoplis, 1983). As product surface temperature elevates, more energy is absorbed by the dry mass in the flake (Fig. 15.4). As the temperature of the dry mass increases, the diffusion rate of water moving through the product increases. In this phase, the amount of water evaporation

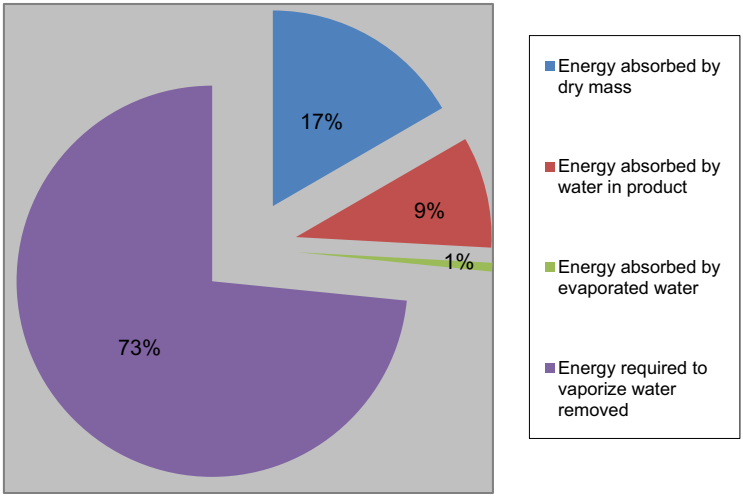

Figure 15.3 Constant rate drying phase: Energy absorbed by product.

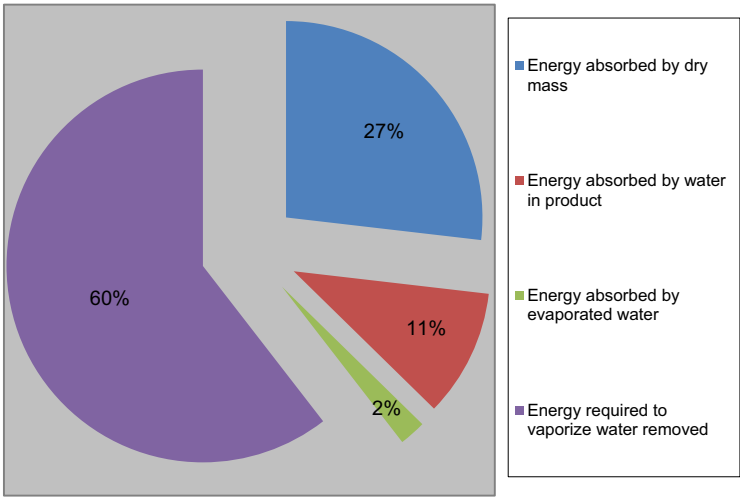

Figure 15.4 Falling rate drying phase: energy absorbed by product.

is impacted by the energy content of the treatment air. This phase ends when the product surface temperature reaches 221°F (105°C) when product starts to change color and expand (see Fig. 15.1). It should be noted that low air energy content (air enthalpy) will extend the falling rate drying phase, whereas high air energy content will shorten the falling rate drying phase.

Expansion/puffing phase

The puffing or expansion phase begins when the product reaches 221°F (105°C) and a change in the physical density occurs. The expansion time is less than 10 s with the moisture content of the flakes dropping quickly from 10.5% to 6.5% wet basis. The puffing phase is marked by two processes (Fig. 15.5). First, the evaporation slows as the water is converted into vapor, which causes the pressure within the starch matrix to increase. At a certain pressure, the starch matrix yields with the product expanding and the vapor is released. Expansion can be the blisters on flakes of corn to the puffing of crispy rice. In general, the amount of expansion is dependent on the rate of temperature increase. For products that do not expand, the puffing zone does not require any additional time or oven length.

Toasting (color and flavor development) phase

The toasting phase starts at the end of the tempering phase and includes Maillard browning and caramelization reactions that change the color and flavor of the flakes. These are kinetic reactions that require an activation energy and then are driven by

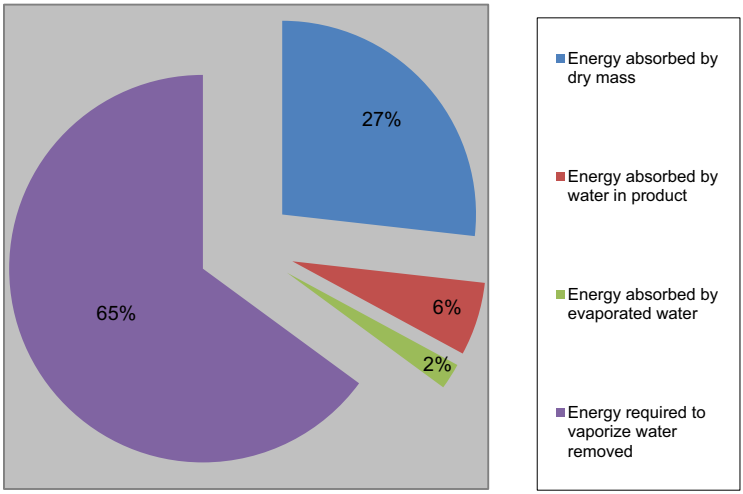

Figure 15.5 Expansion/puffing phase: energy absorbed by product.

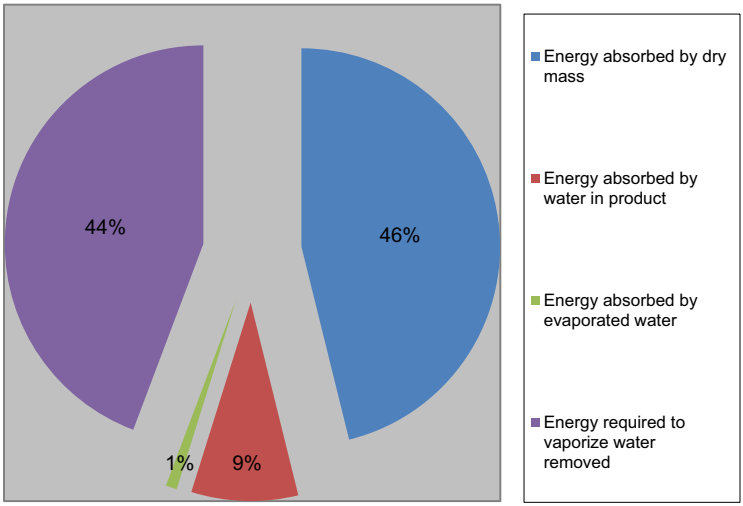

Figure 15.6 Toasting phase: energy absorbed by product.

temperature as the product goes through the oven (Fig. 15.6). Once the reactions start, they continue until the flake temperature falls. The reaction rates are controlled by the amount of energy absorbed by the flake. If these reactions go to completion, flakes will become very dark in color and have bitter flavor notes. Fig. 15.7 illustrates the flake temperature increases slowly, thereby limiting the flavor and color reactions.

The time a product requires in each of the phases depends on product formulation. Shred products entering the oven at a high moisture content will spend a long time in the heating and drying phases. Products with low moisture content will heat up very fast and have a short time in the drying phase.

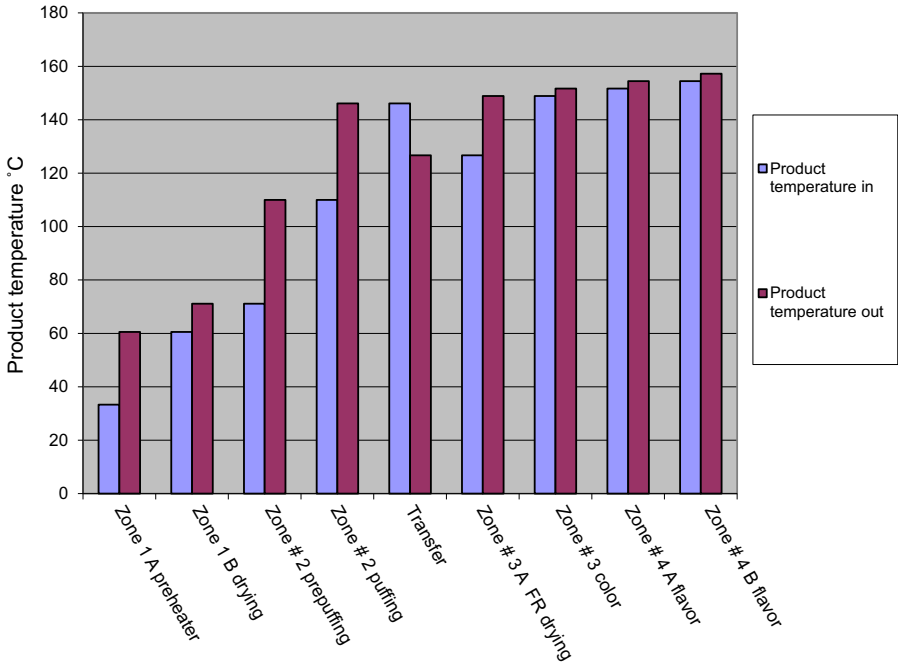

Figure 15.7 Cereal flake temperature during toasting.

Heat and mass transfer equations are used to predict how much of the absorbed energy is used to heat the dry mass and internal water. The basic equations assume that all moisture evaporation takes place at the outer surface. In this way, the moisture transfer becomes a boundary condition for the heating equation. The physical cooling taking place as water vapor evaporates away from the product reduces the surface temperature.

This approach fits well with flakes or products that are very thin and have a small difference in temperature and moisture content between the surface and the core. Inspecting how the energy is absorbed provides insight into how the flakes are physically changing in the toasting process. Figs. 15.7–15.9 illustrate a complex flake product that undergoes a puffing reaction.

Flavor and color changes are affected by many factors, such as moisture content and composition, and occur when the product reaches the activation temperature of 248–284°F (120–140°C) where the Maillard reactions between sugars and amino acids start (Reineccius, 2006). The key control to these reactions is the amount of time spent above the activation temperature and the amount of energy absorbed by the product.

Stopping the chemical reactions is important; the longer the browning reactions continue, the more flavor is generated. Allowed to go to completion, the product will develop a burnt flavor and a dark color. The browning reactions can be stopped by lowering product temperature, e.g., when product exits the toaster.

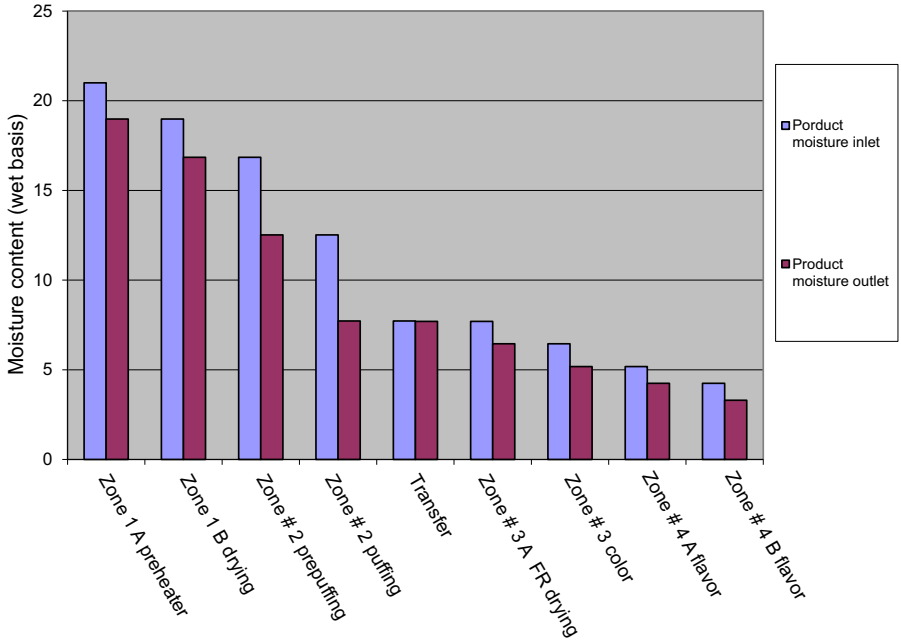

Figure 15.8 Cereal flake moisture content during toasting.

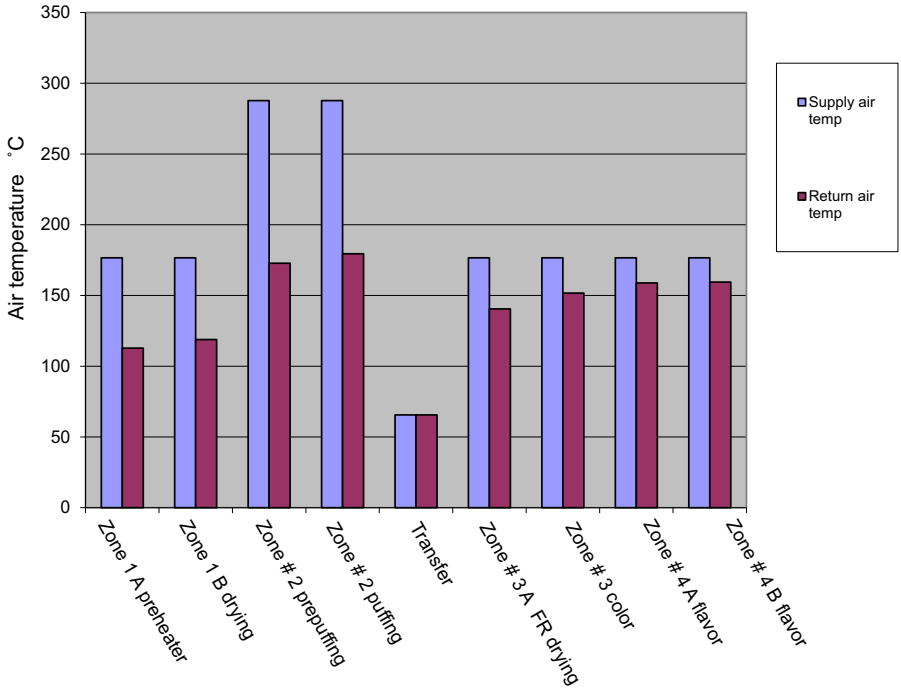

Figure 15.9 Temperature differences between supply and return air at different toasting oven zones.

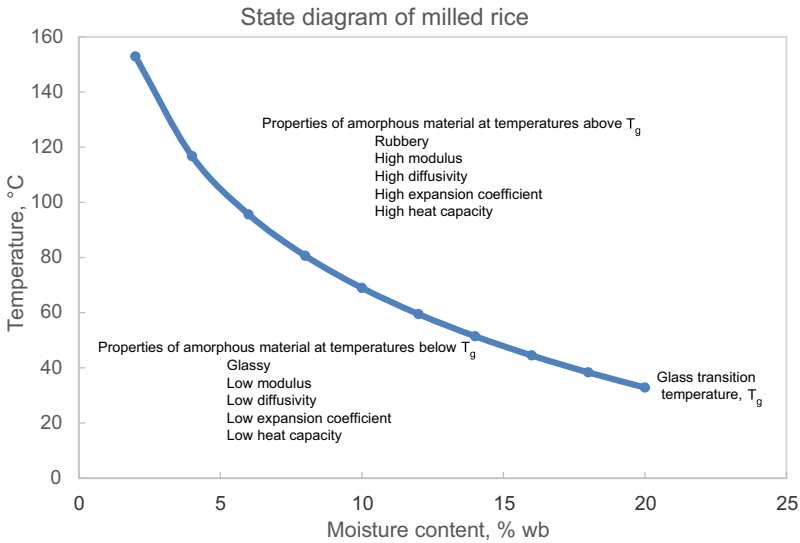

Figure 15.10 Properties of milled rice below and above its glass transition temperature. Adapted from Perdon, A.A., Siebenmorgen, T.J., Mauromoustakos, A., 2000. Glassy state transition and rice drying: development of a brown rice state diagram. Cereal Chem. 77, 708−713.

Products with greater thickness and a defined moisture gradient from the inside to the outside are more complex. As discussed in Chapter 6, product moisture content will affect its T_g. Fig. 15.10 shows the state diagram of milled rice and the differences in its properties above and below T_g (Perdon et al., 2000). Below T_g, the material will be glassy and molecular diffusivity will be slow, whereas above T_g, it will be rubbery and molecular diffusivity will be fast (White and Cakebread, 1966). These material property differences will affect the rate of water evaporation and product temperatures during toasting.

For thick products, the difference in product temperature and moisture content from the outer surface to the product center limits the evaporation and heating occurring. Fast heating of thick product will create an outer boundary or glassy shell that insulates the inner layers, slowing the heating and drying phases. To eliminate this, the air velocity is reduced, and the air humidity is elevated. Thicker products, such as bread or cookies, spend much more time in the heating and constant rate drying phases due to evaporation cooling the outer surface. Water inside the product starts in a liquid state. The surrounding starch matrix exerts pressure on the water, changing how the water molecules move through the starch matrix and elevating the boiling point of the liquid fraction. The water molecules can be in two forms: a liquid fraction and a vapor fraction (Fortes and Okos, 1981). The liquid fraction moves through the starch matrix obeying laws of diffusion. When the liquid fraction comes to a void or pore space within the starch matrix, liquid must be converted into vapor to move across the void. The vapor fraction enters the cooler starch matrix and recondenses into a liquid phase.

Impact of oven type on cereal properties

To understand how the physical and chemical changes are occurring, it is critical that the product be exposed to a uniform treatment from the toasting oven. Inside production ovens, we focus on the product bed, not any individual particle. The most critical parameter inside the toasting oven is the residence time of the product. If the residence time is not tightly controlled, the difference between individual particles becomes larger than the difference between heating phases within the product. The more tightly one can control the residence time, the more tightly one can control the product uniformity.

The easiest residence time control comes when the product is firmly attached to a belt. Baking ovens using this technology provide a very uniform residence time but low heat and mass transfer rates. These low transfer rates are great with thick products like bread and cookies where there are large product temperature and moisture gradients between the center and outer surface. The slow heat transfer of baking ovens does not translate well to traditional breakfast cereal products like flakes or puffed rice that require higher rates of heat transfer.

Fluidized bed ovens present the most difficulty in controlling product residence time. The path the product takes through the oven is impacted by product inlet rates, airflow, conveyor setup, and internal air balance. However, fluidized bed ovens can deliver the high heat and mass transfer required to toast most classical and common cereal products. The high heat and mass transfer rates cause small changes in residence time to make a huge difference in product appearance and uniformity.

There are several styles of fluidized bed ovens. Top-down fluidization provides a more uniform product residence time than other forms of fluidized ovens. Top-down fluidized bed ovens utilize metal tubes to convert air pressure into air velocity at the bottom of the tubes. These tubes act as a resistance to air movement, creating pressure above the tubes. Air always finds the path of least resistance; therefore, the more resistance, the higher the pressure above the tubes and the more uniform the airflow. In this way, fluidized bed ovens provide a uniform air velocity through each tube.

The air velocity created by each tube is focused at the conveyor, as shown in Fig. 15.11. When the air jet hits the conveyor, the air velocity is inverted, into fountain or bowl shape. The air velocity inside this bowl decreases quickly as the air moves up and away from the center of the bowl. High speed air at the center of the bowl is used to lift the product. If the air velocity at any point on the cone is larger than the product's terminal velocity, the air velocity required to lift the product, the product will be lifted and moved in the same direction as the air jet away from the center of the bowl. As the product moves away from the center of the bowl, the air velocity drops. When the air velocity of the cone is less than the terminal velocity of the flakes, the flakes drop back toward the conveyor. When the flakes enter a high velocity region of the cone, the product is again lifted. Flakes move up and down in a circular pattern around the air jet.

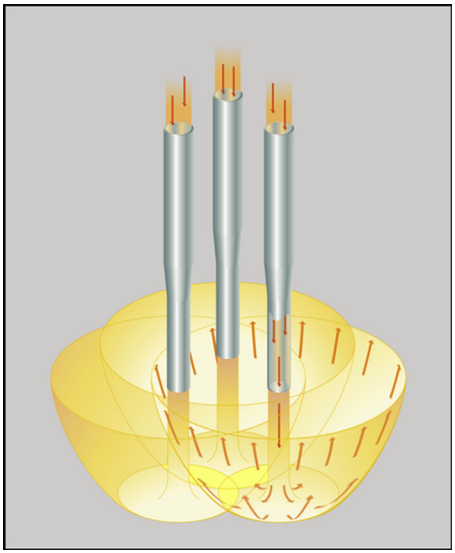

Figure 15.11 When air jets hit the conveyor, the air velocity is inverted into fountain or bowl shape.

When the product bed is fully fluidized, product is suspended in the air stream. At this point, the product bed acts like a liquid: any force acting on the product bed will cause the bed to change or move. The largest force on any product inside the oven should be other product entering the oven pushing the product in the oven toward the exit.

Fluidized bed ovens are very sensitive to uneven loading, which can lead to either more product entering on one side of the oven or the loading changing with time. Both cases will change the product residence time.

Oven internal air movement and initial product velocity are two of the forces that can change the product residence time and shape of the product bed. When air movement influences the product bed, the path the product takes through the oven is changed. Product that takes a longer path takes longer to transit the oven and will result in different product color, density, and moisture content. This can be seen directly with variation of product color across the product bed.

Oven internal air movement issues are often caused by side-to-side air balance. Side-to-side air balance is based on what happens to the air after it passes through the product. As the air passes through the product bed, it loses velocity and is removed from the toaster via the ducts, as shown in Fig. 15.12. The volume of air collected by the duct on each side must be equal; if more air is collected on one side, the product will move across the width of the oven. This results in a longer path through the oven and a nonuniform residence time. Residence time is the most critical variable and any change will impact product image and quality. Side-to-side imbalance is the most common issue during production.

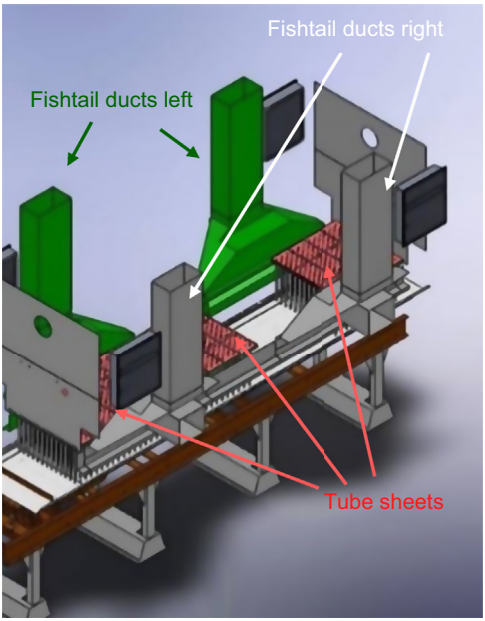

Figure 15.12 The pressurized heated air from the combustion chamber flows through the tube sheets to treat the product. As air passes through the product bed, it loses velocity and is removed via fishtail ducts on either side.

The side-to-side balance can be measured in several ways: by measuring the air volume in each return duct, measuring a difference in humidity from both sides of the oven, or measuring the air temperature inside the ducts. If the air temperature in the duct is higher on one side than the other side, the air balance is not correct, and the product residence time will vary.

The height of the product lift is governed by the pressure above the tubes. Increasing the pressure will increase the height the product is lifted, and the distance a product must move up and down through the oven.

The location of each tube is critical to product fluidization. Each of the air cones must overlap to allow flakes to travel inside the oven without touching the pan. If cones are spaced too far apart, flakes will fall between cones and drop onto the pan. Flakes on the pan have an opportunity to stick to the pan, causing a chain reaction that will start a fire. If cones are packed too tightly, there will be areas in which the velocity of the cones is additive. In these regions, the flakes will be lifted much higher, changing the product residence time.

Product movement in a fluidized bed oven is dominated by the gravitational forces created by the product loading and the elevation drop from inlet to outlet. The slope of the pan controls the distance a product will travel down the oven each time it is lifted up. The mass of new flakes entering the oven acts like a weight pushing the flakes in front and adds to the gravitational force moving the product down the oven.

Types of ovens

This section explores how different oven configurations uniquely match the requirements of specific product types. Additionally, the economics of different ovens will be evaluated in terms of the volume of cereal produced.

The key economical parameter is the volume or mass of toasted cereal produced per day. The capacity of an oven is governed by both the oven type and its physical size. Oven capacity can be categorized by convective heat and mass transfer coefficients: the higher the transfer coefficient, the higher the overall capacity, as illustrated by Fig. 15.13.

All industrial ovens are designed using hot air for heating the product and removing water vapor. Oven design governs how the product and the air meet and interact. Oven operators control air properties to produce flavors and texture attributes required to be successful in the cereal market.

Toasting ovens can be classified by the level of convective heating. The convective heat transfer and mass transfer coefficients are a function of the air properties and air velocity as the air contacts the product surface.

Fluidized bed ovens

Thin products require ovens with the highest convective heat and mass transfer coefficient to produce the largest volume or mass of toasted cereals per unit area. Ovens with fluidized beds offer the highest convective heat and mass transfer coefficients. Fluidized bed ovens suspend the product in the air stream and treat all product surfaces the entire time inside the oven, while maintaining the highest velocity, or difference between the air and product. High velocity air also mixes the product, eliminating gradient in bed depth.

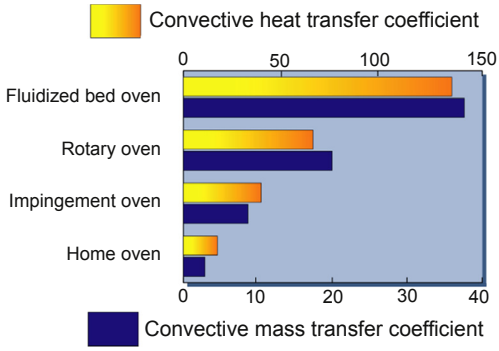

Figure 15.13 Comparison of convective heat transfer and convective mass transfer coefficients in different types of oven.

Product movement within a fluidized bed oven impacts product uniformity. Fluidized bed ovens that are configured with a difference in elevation between the inlet and outlet utilize gravity as the primary force pushing the product down the oven. Fluidized bed ovens that are configured with no elevation difference between the inlet and outlet are utilizing the conveyor as the primary force pushing the product down the oven. Fluidized bed ovens that fluidize the product utilizing top-down air movement are more uniform than ovens that utilize bottom-up fluidization due to impact of air movement (Fig. 15.14).

Design of the air treatment system differentiates fluidized bed manufacturers. Fig. 15.15 shows an example of a fluidized bed oven that maintains the highest Reynolds number, a unitless number that qualifies the level of turbulence of the air moving between the product surface and the air stream, thereby maximizing the product surface area exposed to the air stream.

Impingement ovens

Impingement ovens have lower convective heat and mass transfer coefficients, and therefore offer lower overall capacities per unit volume. These ovens are designed for thick products with large moisture gradients from center to surface.

Rotary ovens

Rotary ovens were the first commercially designed toasting ovens. Rotary ovens utilize mechanical rotation to lift product, which are supported by the outer reel and expose product to airflow from the inner reel.

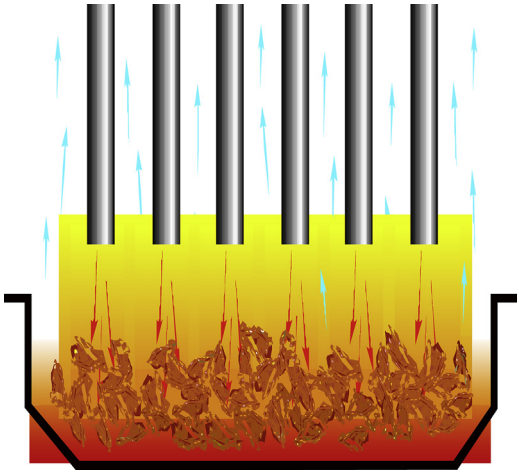

Figure 15.14 Air movement in a fluidized bed oven.

Figure 15.15 Fluidized bed ovens that fluidize the product utilizing top-down air movement are more uniform than ovens that utilize bottom-up fluidization due to impact of air movement. Photo courtesy of Process Partners, Inc.

The path taken by a particle in a rotary oven is the most complex route of any of the various oven types and depends on oven loading, speed of the rotating reel, and the inclination angle of the oven, as detailed in Figs. 15.16 and 15.17.

Air is blown from the inner reel toward the outer reel in all directions. Flakes are lifted by the outer reel using centrifugal force, friction, and air velocity until they reach the point on the drum at which they will either be lifted off the drum and fluidized or

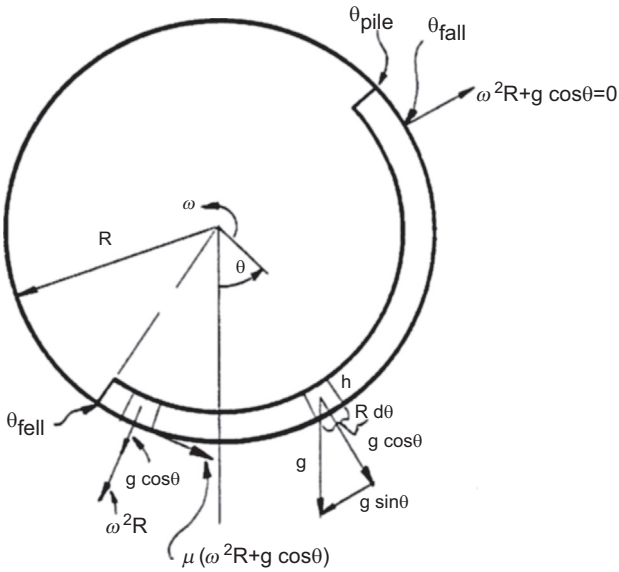

Figure 15.16 Speed of the rotating reel and the inclination angles in a rotary oven.

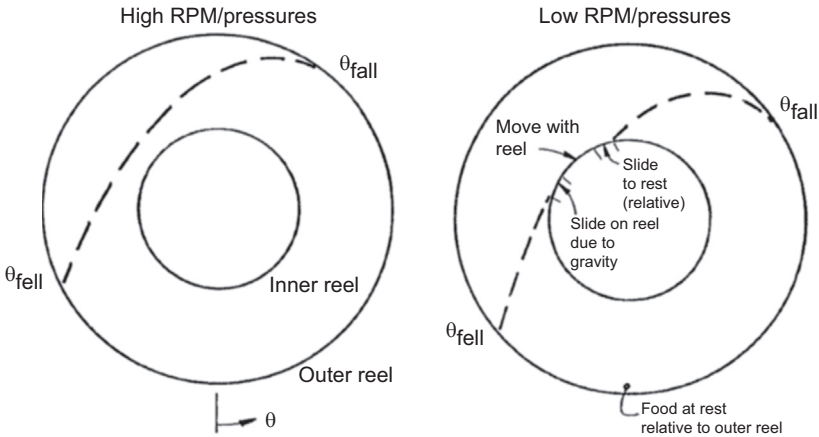

Figure 15.17 Movement path taken by a particle in a rotary oven is the most complex route of any of the various oven types and depends on oven loading, speed of the rotating reel, and the inclination angle of the oven.

tumble back down toward the bottom. The rotation rate is key to the oven design. If the flakes are elevated above the inner reel, they will be fluidized dropping down. One theory is that the flakes form a column on the inside of the drum and stack up until they reach the take-off point.

When the flakes reach the high point of the rotation, gravitational force pulls the flakes off the outer reel and they then fall on a trajectory above the inner reel down to the bottom. The flakes travel countercurrent to the air stream until it passes the reel and then changes direction to flow cocurrently with the air. The speed of the flake increases once it passes the inner reel until it hits the outer reel, where it begins its way up the column of flakes until it reaches the take-off point again. The rotational speed of the outer reel is critical to the flight of the flakes. If the speed is too low, the flake will hit the inner reel and slide off, as shown in Fig. 15.17.

There are two distinct zones of heat and mass transfer in the rotary oven. The first zone is in the falling zone, where the flakes are in contact with the air stream. The second zone is the packed zone where the flakes are in a packed column traveling up the side of the outer reel. Convection dominates the first zone and conduction controls the second zone, where heating and drying are slower. Because convection is the major source of the energy absorbed by the flakes, the design should maximize the time the flake spends in the falling zone.

If flakes impact the inner reel, they are shielded from fluidized convection, which slows the increase in product temperature and decreases the rate of drying, reducing the oven capacity. If the rotation speed is too high, the flakes will hit high on the outer reel, which also limits the time the food spends in the fluidized convection phase and reduces oven capacity.

Gravity conveys the product through the oven. The configuration of the reel shows an elevation drop from the inlet to the outlet. When the flakes are in the fluidized

convection phase, they move a set distance down the drum based on the elevation change. Modification of the reel slope changes the residence time in the oven. Fragile products, such as certain flakes or other pelletized products, shatter into smaller fragments in a rotary oven.

Vapor equilibrium and the thermodynamic cycle

Evaluation of the toasting process highlights two concepts critical to the oven performance, vapor equilibrium and thermodynamic cycle. Vapor equilibrium defines the capacity of the oven, whereas oven design and operating conditions describe a product's thermodynamic cycle, which controls heat requirement, how much heat, and when to hold the product at a constant temperature.

Vapor equilibrium

The art of cereal production matches product appeal with the tools required for mass production. Products are designed to deliver color, flavor, and texture appealing to the consumer. Oven humidity is one of the essential parameters used to control these product attributes. The art of cereal toasting finds and maintains the correct oven humidity, allowing the operator to make the best cereal product, while maximizing production capacity.

The air humidity in the food chamber must be maintained within the correct range for the product. If the supply air dew point temperature (the temperature at which water condenses from the air) is above the product surface temperature, a film of water will condense on the product. The water from the supply air forms a film on the product surface, the surface becomes sticky, products touch each other and form clumps, which then exit the oven untoasted and become a quality defect.

Controlling the air humidity entering the food chamber requires a balance between the water vapor entering the oven and the water vapor leaving the oven. Table 15.1 lists sources of water vapor entering and leaving the oven.

Table 15.1 Sources of water leaving and entering a toasting oven.

Inlet/Outlet	Description
Inlet	Water entering with product
Inlet	Water evaporated from the product
Inlet	Water produced in the combustion process
Inlet	Water vapor coming in with makeup air
Inlet	Water vapor entering from the room
Outlet	Water vapor removed as exhaust
Outlet	Water leaving with the product

An oven is not a closed system. As air is exhausted from the oven, makeup air is pulled into the oven. In steady-state operating conditions, the water vapor entering the oven is balanced by water vapor leaving in the exhaust air.

If the exhaust system is designed correctly, the water vapor entering or generated in the oven is totally removed by the exhaust system. If the exhaust is correct, the mass of air leaving with the exhaust is balanced by makeup air entering. Makeup air enters the oven at the open ends, where food moves into and out of the oven, or through specific makeup ducts. The makeup system is designed to balance the mass of air removed in the exhaust.

Typical oven mass balance

Tracking the air as it circulates through an oven illustrates how the humidity changes (Table 15.2).

1. As the air comes in contact with the food, the humidity must be controlled within tight limits, to develop the desired taste and texture of the product. As the air passes through the product bed, water is evaporated from the product, and water vapor from the room mixes with the circulation air resulting in a drop in air temperature and increase in humidity.
2. Air at the elevated humidity is removed by the exhaust system, lowering the mass of water vapor left inside the oven. The exhaust flow does not lower the humidity level but does reduce the mass flow.
3. Makeup or room air is added to the circulation air, replacing the wet air removed in the exhaust, and with the makeup air comes water vapor from the room humidity.
4. The circulation air is heated by the burner, which contributes water vapor from the combustion process to the air. The heated air, now at the correct humidity, comes back into contact with the food product. The air humidity at the exhaust sets the amount of water removed with each unit volume of air. The mass of air per unit time requiring exhaust can be calculated by

Table 15.2 Changes in air humidity as it circulates through an oven.

Drying air	Air mass (lb/min)	Water mass (lb/min)	Mixture	Humidity ratio
Inlet toasting air	645.5	48.4	693.9	0.075
Entering from room	32.0	0.5	32.5	0.016
Product water vapor	n/a	20.5	20.5	n/a
Removed by exhaust	−295.3	−30.2	−325.5	0.102
Makeup air	216.2	3.3	219.5	0.015
Combustion air	47.0	0.7	47.7	0.015
Water produced by combustion	0.0	5.3	5.3	n/a
Net	645.5	48.4	693.9	0.075

dividing the amount of water entering the oven by humidity at exhaust point. The air mass can be converted into a volume per unit time by dividing the mass of moist air by its density.

Air properties can be found on a psychrometric chart. Fig. 15.18 details changes in air properties inside the oven. As manufacturers increase the capacity of an oven, the mass of water being evaporated from the product increases. This increases the demands on the exhaust system and increases the volume of air being exhausted.

When the physical size of the exhaust duct is not sufficient to remove all of the water entering the system, the humidity inside the oven will increase. This may slow the rate of drying and limit the capacity of the oven. This is the first capacity that limits most ovens. Operators will search for new ways to get the moisture back under control. Pressure changes inside the oven affects food residence time, resulting in spotty food due to variations in color and moisture content. Spotty color is measured as a quality defect: one quality problem has been traded for a second quality defect. Operators will work back and forth between the two quality defects of high moisture and spotty color, unable to make the system run correctly. This is a typical design problem. The exhaust system must be designed to remove the required water vapor.

Toasting ovens operate at temperatures in excess of 600°F (315°C). The density of air at this temperature is half of the density of air under standard conditions, 70°F (21°C), with 50% relative humidity at sea level. As the temperature of the air rises, the density falls and the required duct size increases.

As the air picks up water vapor, its density drops because water vapor is lighter than air. As the moisture load increases, the air density goes down, increasing the required size of the exhaust duct.

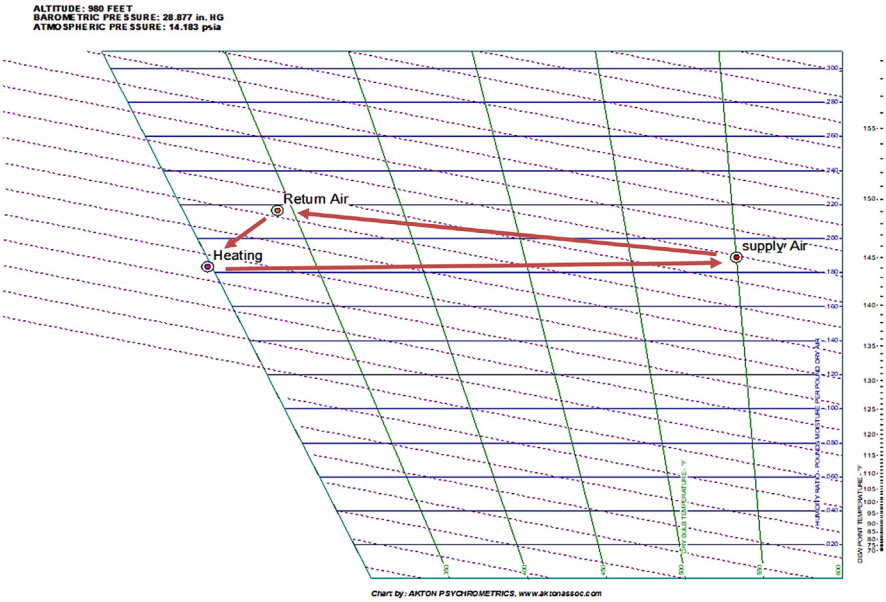

Figure 15.18 Psychrometric chart detailing change in oven air properties.

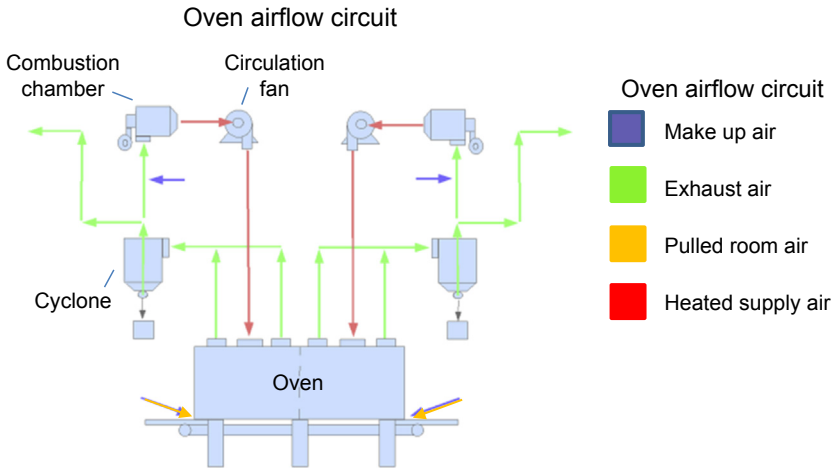

Figure 15.19 Airflow path in an oven.

Applying the knowledge about the product path and oven operating conditions provides a method to look at how the product changes inside the oven. This insight can be used to improve the oven design and operation. Conductive and convective heat transfer (Fig. 15.20) impacts product temperature during toasting. Reviewing the temperature changes that occur in the product is a good method for understanding toasting, as shown in Fig. 15.1.

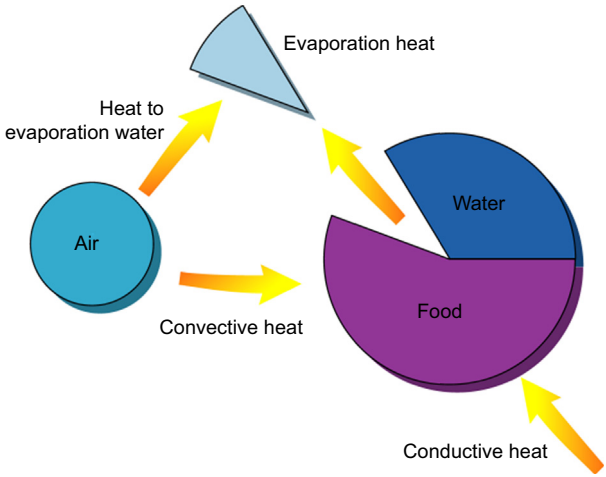

Figure 15.20 Conductive and convective heat transfer process.

Toasting sensitivity studies

Toasting of cereals is the final step in a complex production process. Process variations in each step of the production system are carried into the toasting operation. Given all the variation coming into the oven, toasting is required to produce a uniform product.

Variations that make a large change in the product output or the performance of the toasting operation are described as sensitive. Sensitive variables can be identified by making a small change in a variable and evaluating the oven output. Simulations can be used to evaluate the sensitivity of variables that are inputs to the toasting process, or variables that have not been identified by the art as part of this category.

The art of toasting cereals has defined many of the most sensitive parameters. These are the variables that have evolved into control systems. The oven temperature, the airflow rates, and the residence time are examples of sensitive parameters that are under the control of the operator (Figs. 15.21–15.23).

For all types of toasting ovens, the initial moisture content is sensitive. Fig. 15.21 demonstrates what changes in initial moisture content the oven can absorb without changing the product. Statistically, there will always be a natural variation in the moisture content of product coming into the oven. This figure demonstrates how many standard deviations in the moisture can change before the quality of the product is changed.

If the oven is designed correctly, the sensitivity to initial moisture content can be minimized. This becomes an additional phase of the toasting cycle. This tempering phase is after the drying section. The tempering phase has a dry bulb temperature below the product temperature. When the product enters the tempering phase, the high moisture content product continues to dry. The low moisture content product is

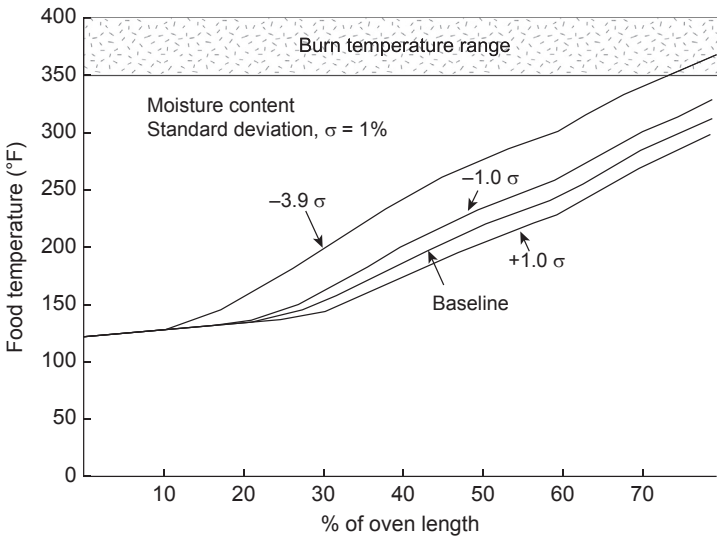

Figure 15.21 Effect of initial moisture content on food temperature versus time.

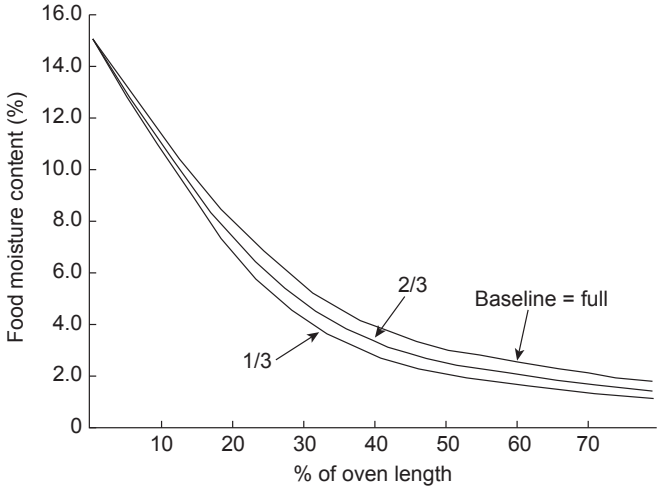

Figure 15.22 Effect of one-third and two-thirds load reduction on food moisture versus time.

held at the wet bulb and suspends the drying process. The result is all products enter the toasting phase at the same temperature and moisture content.

In production, changes of the food load into the oven can move up and down. This can be observed in a flaked product when a mill is taken off line. The operators must make changes in the oven operation to compensate for this load change. Fig. 15.2 shows changes in moisture content in the toasting phase as load changes. When the load into the oven drops, without compensation in the toasting parameters, the product temperature changes, as seen in Fig. 15.23. The temperature history makes a large jump and must be compensated for by lowering the air temperature.

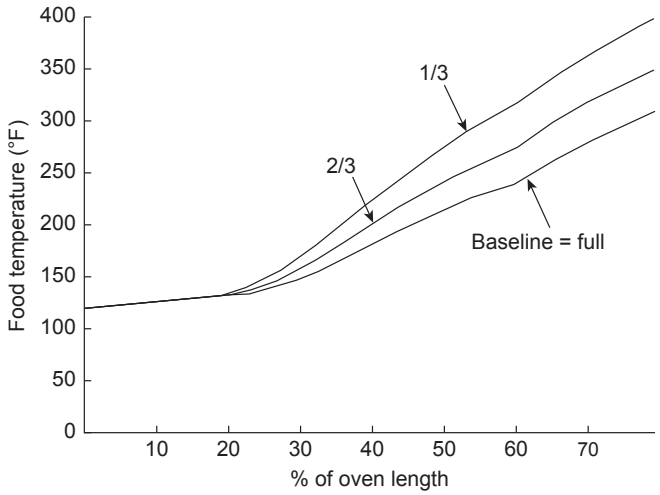

Figure 15.23 Effect of one-third and two-thirds load reduction on food temperature versus time.

Summary

Advancements in toasting started in the baking industry and have been refined and expanded by the breakfast cereal industry. Toasting is critical to the manufacture of breakfast cereals because it sets the final attributes consumers find most appealing, such as flavor, color, and texture. Fundamental understanding of product expansion and flavor development thermodynamics can be applied to improve flavor profile and texture of products. Improving the oven efficiency, or capacity, and product uniformity requires understanding energy balance and how best to configure the oven to meet product requirements.

References

Fortes, M., Okos, M.R., 1981. A non-equilibrium thermodynamics approach to transport phenomena in capillary porous media. Trans. ASAE 24 (3), 756–760.

Geankoplis, C.J., 1983. Transport Processes and Unit Operations, third ed. Prentice Hall, Englewood Cliffs, NJ 07632, pp. 539–549.

Perdon, A.A., Siebenmorgen, T.J., Mauromoustakos, A., 2000. Glassy state transition and rice drying: development of a brown rice state diagram. Cereal Chem 77, 708–713.

Reineccius, G., 2006. Flavor Chemistry and Technology. CRC Press/Taylor & Francis, Boca Raton, FL 33487, pp. 103–110.

White, G., Cakebread, S., 1966. The glassy state in certain sugar-containing food products. Int. J. Food Sci. Technol 1, 73–82.